

Smart Shop Review System

E-commerce platforms like Amazon and Flipkart generate millions of product reviews, images, and demo videos every day. However, a significant portion of these reviews are fake, biased, or bot-generated, making it extremely difficult for buyers to trust the information available to them. Shoppers are forced to manually sift through hundreds of text reviews, blurry product images, and unverified demo videos before making a purchase decision — a process that is both time-consuming and unreliable. This results in poor buying choices, increased product return rates, and declining consumer confidence in online marketplaces.

You are tasked with building a Machine Learning-powered Smart Shop Review System that collects multimodal product data — product images scraped from Amazon and Flipkart, demo videos downloaded from YouTube, and customer text reviews extracted using BeautifulSoup or Selenium — and applies ML techniques to analyze each data type. The system must use Computer Vision to assess image quality, Natural Language Processing (NLP) to perform sentiment analysis on text reviews, and video frame analysis to extract meaningful signals from demo videos. The goal is to generate a unified Smart Review Score for each product, answering the following key question:

“How can machine learning models trained on product images, demo videos, and customer text reviews be combined to automatically detect fake reviews, assess product quality, and generate a reliable Smart Review Score that helps online shoppers make confident and informed purchase decisions?”

Deliverables

- 1. Data Collection Pipeline (Python — Web Scraping):** Scrape product images from Amazon and Flipkart using BeautifulSoup and Selenium. Download product demo videos from YouTube using yt-dlp or the YouTube Data API. Extract customer text reviews from product listing pages and store all data in a structured format for ML processing.
- 2. Image Processing Module (Computer Vision — CNN):** Apply Convolutional Neural Networks (CNN) or pre-trained models such as ResNet and MobileNet to classify product image quality, detect misleading visuals, and extract visual features. Flag products where images do not match customer complaints or review sentiments.
- 3. Text Review Analysis Module (NLP — Sentiment Analysis):** Preprocess and clean raw customer review text. Train or fine-tune NLP models (such as LSTM, BERT,

or logistic regression with TF-IDF) to classify reviews as Positive, Negative, or Neutral. Implement a fake review detection classifier using anomaly detection or supervised ML techniques.

4. Video Analysis Module (Frame Extraction & Classification): Extract keyframes from product demo videos and apply image classification models to assess whether demo content accurately represents the product. Identify misleading or low-quality demo videos using frame-level feature analysis.

5. Smart Review Score Engine & Dashboard (Python — Streamlit): Combine signals from image quality, sentiment scores, fake review flags, and video analysis into a single weighted Smart Review Score per product. Build an interactive Streamlit dashboard displaying scores, sentiment breakdowns, and image/video quality flags for each product.

6. Report and Presentation: Write a concise project report covering data collection methodology, ML model architectures, evaluation metrics, and key findings. Prepare a presentation communicating insights and business recommendations to stakeholders.

7. GitHub Repository: Include all Python scripts, scraping code, trained ML models, Jupyter notebooks, and dashboard files in a well-structured and fully documented repository.